

# Advanced Computer Networks – PCS223

## ASSIGNMENT 2

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Subgroup – CS2

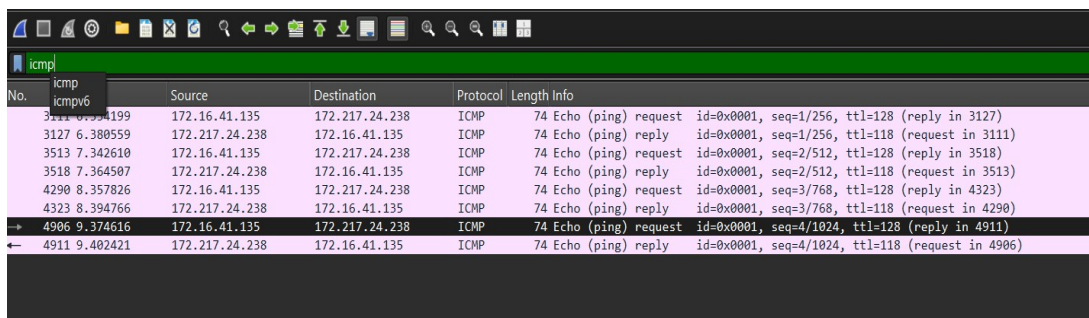
Roll No. – 8025320034

### 1. To capture and analyze ICMP packets (Ping request and reply)

```
C:\Users\dell9>ping google.com

Pinging google.com [172.217.24.238] with 32 bytes of data:
Reply from 172.217.24.238: bytes=32 time=46ms TTL=118
Reply from 172.217.24.238: bytes=32 time=22ms TTL=118
Reply from 172.217.24.238: bytes=32 time=37ms TTL=118
Reply from 172.217.24.238: bytes=32 time=27ms TTL=118

Ping statistics for 172.217.24.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 46ms, Average = 33ms
```



| No.    | icmp<br>icmpv6 | Source         | Destination    | Protocol | Length | Info                                                               |
|--------|----------------|----------------|----------------|----------|--------|--------------------------------------------------------------------|
| 3111   | 6.354199       | 172.16.41.135  | 172.217.24.238 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 3127)  |
| 3127   | 6.380559       | 172.217.24.238 | 172.16.41.135  | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=1/256, ttl=118 (request in 3111)  |
| 3513   | 7.342610       | 172.16.41.135  | 172.217.24.238 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 3518)  |
| 3518   | 7.364507       | 172.217.24.238 | 172.16.41.135  | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=2/512, ttl=118 (request in 3513)  |
| 4290   | 8.357826       | 172.16.41.135  | 172.217.24.238 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 4323)  |
| 4323   | 8.394766       | 172.217.24.238 | 172.16.41.135  | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=3/768, ttl=118 (request in 4290)  |
| → 4906 | 9.374616       | 172.16.41.135  | 172.217.24.238 | ICMP     | 74     | Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 4911) |
| ← 4911 | 9.402421       | 172.217.24.238 | 172.16.41.135  | ICMP     | 74     | Echo (ping) reply id=0x0001, seq=4/1024, ttl=118 (request in 4906) |

## 2. To analyze TCP packets and observe the TCP three-way handshake

| No.  | Time      | Source         | Destination    | Protocol | Length | Info                                                                                     |
|------|-----------|----------------|----------------|----------|--------|------------------------------------------------------------------------------------------|
| 6892 | 13.725480 | 172.16.41.135  | 13.204.106.34  | TCP      | 55     | 60059 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1                                              |
| 6896 | 13.728040 | 13.204.106.34  | 172.16.41.135  | TCP      | 66     | 443 → 60059 [ACK] Seq=1 Ack=2 Win=103 Len=0 SLE=1 SRE=2                                  |
| 7705 | 14.957805 | 172.16.41.135  | 74.125.200.188 | TCP      | 55     | 58464 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1                                             |
| 7746 | 15.038603 | 74.125.200.188 | 172.16.41.135  | TCP      | 66     | 5228 → 58464 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2                                |
| 8973 | 17.342393 | 172.16.41.135  | 52.111.224.7   | TCP      | 66     | 56569 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM                        |
| 8976 | 17.345076 | 52.111.224.7   | 172.16.41.135  | TCP      | 66     | 443 → 56569 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256             |
| 8977 | 17.345148 | 172.16.41.135  | 52.111.224.7   | TCP      | 54     | 56569 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0                                            |
| 8979 | 17.346649 | 172.16.41.135  | 52.111.224.7   | TLsv1.2  | 250    | Client Hello (SNI=messaging.engagement.office.com)                                       |
| 8983 | 17.349960 | 52.111.224.7   | 172.16.41.135  | TCP      | 60     | 443 → 56569 [ACK] Seq=1 Ack=197 Win=15872 Len=0                                          |
| 9196 | 17.701531 | 52.111.224.7   | 172.16.41.135  | TLsv1.2  | 1514   | Server Hello                                                                             |
| 9197 | 17.701531 | 52.111.224.7   | 172.16.41.135  | TCP      | 1490   | 443 → 56569 [PSH, ACK] Seq=1461 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 9203] |
| 9198 | 17.701630 | 172.16.41.135  | 52.111.224.7   | TCP      | 54     | 56569 → 443 [ACK] Seq=197 Ack=2897 Win=65280 Len=0                                       |
| 9200 | 17.702894 | 52.111.224.7   | 172.16.41.135  | TCP      | 1514   | 443 → 56569 [ACK] Seq=2897 Ack=197 Win=15872 Len=1460 [TCP PDU reassembled in 9203]      |
| 9201 | 17.702894 | 52.111.224.7   | 172.16.41.135  | TCP      | 1490   | 443 → 56569 [PSH, ACK] Seq=4357 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 9203] |
| 9203 | 17.702894 | 52.111.224.7   | 172.16.41.135  | TLsv1.2  | 1502   | Certificate                                                                              |
| 9204 | 17.702894 | 52.111.224.7   | 172.16.41.135  | TLsv1.2  | 1514   | Certificate Status                                                                       |
| 9205 | 17.702894 | 52.111.224.7   | 172.16.41.135  | TLsv1.2  | 362    | Server Key Exchange, Server Hello Done                                                   |
| 9206 | 17.702961 | 172.16.41.135  | 52.111.224.7   | TCP      | 54     | 56569 → 443 [ACK] Seq=197 Ack=5793 Win=65280 Len=0                                       |
| 9207 | 17.703110 | 172.16.41.135  | 52.111.224.7   | TCP      | 54     | 56569 → 443 [ACK] Seq=197 Ack=9009 Win=65280 Len=0                                       |
| 9208 | 17.706135 | 172.16.41.135  | 52.111.224.7   | TLsv1.2  | 180    | Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message                     |
| 9209 | 17.708115 | 52.111.224.7   | 172.16.41.135  | TCP      | 60     | 443 → 56569 [ACK] Seq=9009 Ack=323 Win=15872 Len=0                                       |
| 9363 | 17.883072 | 52.111.224.7   | 172.16.41.135  | TLsv1.2  | 328    | New Session Ticket, Change Cipher Spec, Encrypted Handshake Message                      |
| 9372 | 17.891479 | 172.16.41.135  | 52.111.224.7   | TLsv1.2  | 2313   | Application Data                                                                         |
| 9374 | 17.901531 | 52.111.224.7   | 172.16.41.135  | TCP      | 60     | 443 → 56569 [ACK] Seq=9203 Ack=1703 Win=10600 Len=0                                      |
| 9375 | 17.901531 | 52.111.224.7   | 172.16.41.135  | TCP      | 60     | 443 → 56569 [ACK] Seq=9283 Ack=2582 Win=21760 Len=0                                      |
| 9451 | 18.140473 | 172.16.41.135  | 52.110.14.146  | TCP      | 66     | 56570 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM                        |
| 9452 | 18.140826 | 172.16.41.135  | 52.110.14.146  | TCP      | 66     | 56571 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM                        |
| 9454 | 18.146720 | 52.110.14.146  | 172.16.41.135  | TCP      | 66     | 443 → 56570 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256             |
| 9455 | 18.146720 | 52.110.14.146  | 172.16.41.135  | TCP      | 66     | 443 → 56571 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256             |

|                                                                                                                                                                                                   |                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Wireshark - Packet 8976 - Wi-Fi                                                                                                                                                                   |                                                              |
| ▶ Frame 8976: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0                                             |                                                              |
| ▶ Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60)                                                                                                |                                                              |
| ▶ Internet Protocol Version 4, Src: 52.111.224.7, Dst: 172.16.41.135                                                                                                                              |                                                              |
| ▶ Transmission Control Protocol, Src Port: 443, Dst Port: 56569, Seq: 0, Ack: 1, Len: 0                                                                                                           |                                                              |
| Source Port: 443                                                                                                                                                                                  |                                                              |
| Destination Port: 56569                                                                                                                                                                           |                                                              |
| [Stream index: 2]                                                                                                                                                                                 |                                                              |
| [Stream Packet Number: 2]                                                                                                                                                                         |                                                              |
| ▶ [Conversation completeness: Complete, WITH_DATA (31)]                                                                                                                                           |                                                              |
| [TCP Segment Len: 0]                                                                                                                                                                              |                                                              |
| Sequence Number: 0 (relative sequence number)                                                                                                                                                     |                                                              |
| Sequence Number (raw): 1333957722                                                                                                                                                                 |                                                              |
| [Next Sequence Number: 1 (relative sequence number)]                                                                                                                                              |                                                              |
| Acknowledgment Number: 1 (relative ack number)                                                                                                                                                    |                                                              |
| Acknowledgment number (raw): 1272899674                                                                                                                                                           |                                                              |
| 1000 ... = Header Length: 32 bytes (8)                                                                                                                                                            |                                                              |
| ▶ [Flags: 0x012 (SYN, ACK)]                                                                                                                                                                       |                                                              |
| Window: 14600                                                                                                                                                                                     |                                                              |
| [Calculated window size: 14600]                                                                                                                                                                   |                                                              |
| Checksum: 0x551f [unverified]                                                                                                                                                                     |                                                              |
| [Checksum Status: Unverified]                                                                                                                                                                     |                                                              |
| Urgent Pointer: 0                                                                                                                                                                                 |                                                              |
| ▶ Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window scale                                                             |                                                              |
| ▶ [Timestamps]                                                                                                                                                                                    |                                                              |
| ▶ [SEQ/ACK analysis]                                                                                                                                                                              |                                                              |
| [Client Contiguous Streams: 1]                                                                                                                                                                    |                                                              |
| [Server Contiguous Streams: 1]                                                                                                                                                                    |                                                              |
| 0000                                                                                                                                                                                              | 98 43 fa 0a fe 60 04 d5 90 6e 91 0f 08 00 45 00 .C...n...E   |
| 0010                                                                                                                                                                                              | 00 34 f9 1f 40 00 40 06 57 96 34 6f e0 07 ac 10 .4.@@W4o...  |
| 0020                                                                                                                                                                                              | 29 87 01 bb dc f9 4f 82 94 5a 4b de e8 5a 80 12 ).---O.ZK.Z. |
| 0030                                                                                                                                                                                              | 39 08 55 1f 00 00 02 04 05 b4 01 01 04 02 01 03 9U-----      |
| 0040                                                                                                                                                                                              | 03 08 ..                                                     |
| No.: 8976 - Time: 17.345076 - Source: 52.111.224.7 - Destination: 172.16.41.135 - Protocol: TCP - Length: 66 - Info: 443 → 56569 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256 |                                                              |

### 3. To capture and analyze UDP packets

| No.  | Time      | Source                        | Destination    | Protocol | Length | Info                                                                                                                            |
|------|-----------|-------------------------------|----------------|----------|--------|---------------------------------------------------------------------------------------------------------------------------------|
| 8968 | 17.321967 | 172.16.41.135                 | 172.31.1.6     | DNS      | 91     | Standard query 0x8464 AAAA messaging.engagement.office.com                                                                      |
| 8969 | 17.323584 | 172.31.1.6                    | 172.16.41.135  | DNS      | 182    | Standard query response 0x1d09 A messaging.engagement.office.com CNAME prod-campaignaggregator.omexexternalfb.office.net.akad-  |
| 8970 | 17.331977 | 173.194.154.38                | 172.16.41.135  | UDP      | 1296   | 443 → 51085 Len=1250                                                                                                            |
| 8972 | 17.337200 | 172.31.1.6                    | 172.16.41.135  | DNS      | 194    | Standard query response 0x8464 AAAA messaging.engagement.office.com CNAME prod-campaignaggregator.omexexternalfb.office.net.a-  |
| 8974 | 17.344875 | 142.250.195.46                | 172.16.41.135  | UDP      | 70     | 443 → 59855 Len=24                                                                                                              |
| 8975 | 17.345072 | 172.16.41.135                 | 142.250.195.46 | UDP      | 73     | 59855 → 443 Len=31                                                                                                              |
| 8978 | 17.345165 | 172.16.41.135                 | 173.194.154.38 | UDP      | 75     | 51085 → 443 Len=33                                                                                                              |
| 8980 | 17.347519 | fe80::aa1:89ff:fe55::ff02::fb | ff02::fb       | MDNS     | 424    | Standard query 0x0000 ANY HIKVISION DS-2CD3651G0-IZ - F60580171._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3651G0-IZ - |
| 8982 | 17.347808 | 172.16.33.33                  | 224.0.0.251    | MDNS     | 400    | Standard query 0x0000 ANY HIKVISION DS-2CD3651G0-IZ - F60580171._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3651G0-IZ - |
| 8986 | 17.382648 | 172.16.33.82                  | 224.0.0.251    | MDNS     | 429    | Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459834._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU - |
| 8987 | 17.383073 | fe80::2628:fdfc:feff::fb      | ff02::fb       | MDNS     | 453    | Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459834._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU - |
| 8988 | 17.388697 | 142.250.195.46                | 172.16.41.135  | UDP      | 114    | 443 → 59855 Len=68                                                                                                              |
| 8989 | 17.397498 | 142.250.195.46                | 172.16.41.135  | UDP      | 67     | 443 → 59855 Len=21                                                                                                              |
| 8990 | 17.397831 | 172.16.41.135                 | 142.250.195.46 | UDP      | 77     | 59855 → 443 Len=35                                                                                                              |
| 8991 | 17.412490 | fe80::2628:fdfc:fecc::fb      | ff02::fb       | MDNS     | 424    | Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31138577._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU - |
| 8992 | 17.412668 | 172.16.33.79                  | 224.0.0.251    | MDNS     | 400    | Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31138577._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU - |
| 8993 | 17.413591 | 173.194.154.38                | 172.16.41.135  | UDP      | 1296   | 443 → 51085 Len=1250                                                                                                            |
| 8994 | 17.418481 | 172.16.36.131                 | 224.0.0.251    | MDNS     | 349    | Standard query response 0x0000 PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local- |
| 8995 | 17.419113 | fe80::1805:a078:e19::fb       | ff02::fb       | MDNS     | 373    | Standard query response 0x0000 PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local- |
| 8996 | 17.419690 | 172.16.36.119                 | 224.0.0.251    | MDNS     | 1047   | Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question PTR Akshin's iPad (2- |
| 8997 | 17.421600 | fe80::1c52:6de9:a09::fb       | ff02::fb       | MDNS     | 1071   | Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question PTR Akshin's iPad (2- |
| 8998 | 17.423066 | 172.16.33.86                  | 224.0.0.251    | MDNS     | 429    | Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459843._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU - |
| 8999 | 17.423962 | fe80::2628:fdfc:feff::fb      | ff02::fb       | MDNS     | 453    | Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459843._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU - |
| 9000 | 17.426562 | fe80::2628:fdfc:feff::fb      | ff02::fb       | MDNS     | 453    | Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459850._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU - |
| 9001 | 17.426789 | 172.16.33.80                  | 224.0.0.251    | MDNS     | 429    | Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459850._http._tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU - |
| 9002 | 17.428458 | 172.16.41.135                 | 173.194.154.38 | UDP      | 75     | 51085 → 443 Len=33                                                                                                              |

```
▶ Frame 8989: Packet, 67 bytes on wire (536 bits), 67 bytes captured (536 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0
  ▾ Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
    ▸ Destination: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
    ▸ Source: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
    Type: 802.1Q Virtual LAN (0x8100)
    [Stream index: 4]
  ▾ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
    000. .... = Priority: Best Effort (default) (0)
    ...0 .... = DEI: Ineligible
    .... 0000 0000 0000 = ID: 0
    Type: IPv4 (0x0800)
  ▸ Internet Protocol Version 4, Src: 142.250.195.46, Dst: 172.16.41.135
  ▸ User Datagram Protocol, Src Port: 443, Dst Port: 59855
    Source Port: 443
    Destination Port: 59855
    Length: 29
    Checksum: 0xe883 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 368]
    [Stream Packet Number: 15]
    [Timestamps]
    ▸ UDP payload (21 bytes)
  ▸ Data (21 bytes)

0000  98 43 fa 0a fe 60 04 d5  90 6e 91 0f 81 00 00 00  .C...n....
0010  08 00 45 80 00 31 00 00  40 00 3c 11 16 7c 8e fa  -E 1...@<|.
0020  c3 2a ac 10 29 87 01 bb  e9 cf 00 1d e8 83 50 1a  (...)...P.
0030  82 10 87 bc 1e e5 46 88  63 93 23 f6 2d 21 0a 13  ....F.c#-!..
0040  e9 d1 9c                                     ....

No: 8989 - Time: 17.397498 - Source: 142.250.195.46 - Destination: 172.16.41.135 - Protocol: UDP - Length: 67 - Info: 443 → 59855 Len=21
```

## 4. To capture and analyze DNS query and response packets

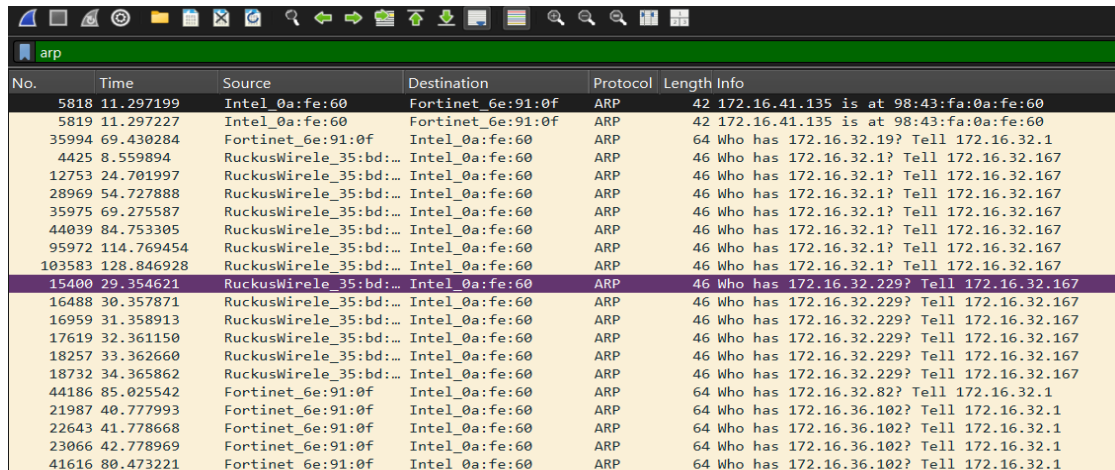
| dns   |           |               |             |          |                                                                |
|-------|-----------|---------------|-------------|----------|----------------------------------------------------------------|
| No.   | Time      | Source        | Destination | Protocol | Length Info                                                    |
| 71950 | 92.027577 | 172.16.41.135 | 172.31.1.6  | DNS      | 92 Standard query 0x0003 A fa000000191.resources.office.net    |
| 16846 | 31.105719 | 172.16.41.135 | 172.31.1.6  | DNS      | 85 Standard query 0x03bf A fd.api.iris.microsoft.com           |
| 75735 | 92.889790 | 172.16.41.135 | 172.31.1.6  | DNS      | 92 Standard query 0x04e6 A fa000000193.resources.office.net    |
| 75773 | 92.908395 | 172.16.41.135 | 8.8.8.8     | DNS      | 92 Standard query 0x04e6 A fa000000193.resources.office.net    |
| 17123 | 31.660998 | 172.16.41.135 | 172.31.1.6  | DNS      | 78 Standard query 0x0ec8 AAAA uci.cdn.office.net               |
| 64804 | 90.185509 | 172.16.41.135 | 172.31.1.6  | DNS      | 92 Standard query 0x0eed A fa000000145.resources.office.net    |
| 66245 | 90.652437 | 172.16.41.135 | 172.31.1.6  | DNS      | 92 Standard query 0x15eb A fa000000175.resources.office.net    |
| 8967  | 17.321777 | 172.16.41.135 | 172.31.1.6  | DNS      | 91 Standard query 0x1d09 A messaging.engagement.office.com     |
| 45655 | 85.668091 | 172.16.41.135 | 172.31.1.6  | DNS      | 92 Standard query 0x278d A fa000000128.resources.office.net    |
| 54196 | 87.491990 | 172.16.41.135 | 172.31.1.6  | DNS      | 92 Standard query 0x293c A fa000000136.resources.office.net    |
| 44264 | 85.236578 | 172.16.41.135 | 172.31.1.6  | DNS      | 92 Standard query 0x2c65 A fa000000124.resources.office.net    |
| 30496 | 57.952915 | 172.16.41.135 | 172.31.1.6  | DNS      | 87 Standard query 0x2dd3 AAAA roaming.svc.cloud.microsoft      |
| 30521 | 57.983417 | 172.16.41.135 | 8.8.8.8     | DNS      | 87 Standard query 0x2dd3 AAAA roaming.svc.cloud.microsoft      |
| 14265 | 27.005871 | 172.16.41.135 | 172.31.1.6  | DNS      | 90 Standard query 0x2e5c AAAA self.events.data.microsoft.com   |
| 14264 | 27.005784 | 172.16.41.135 | 172.31.1.6  | DNS      | 90 Standard query 0x31bd A self.events.data.microsoft.com      |
| 9492  | 18.257844 | 172.16.41.135 | 172.31.1.6  | DNS      | 80 Standard query 0x32c2 A substrate.office.com                |
| 42619 | 82.146241 | 172.16.41.135 | 172.31.1.6  | DNS      | 80 Standard query 0x3535 AAAA cs.dds.microsoft.com             |
| 13718 | 26.441137 | 172.16.41.135 | 172.31.1.6  | DNS      | 76 Standard query 0x391e A fs.microsoft.com                    |
| 66257 | 90.654212 | 172.16.41.135 | 172.31.1.6  | DNS      | 92 Standard query 0x3c3a AAAA fa000000156.resources.office.net |
| 66249 | 90.652830 | 172.16.41.135 | 172.31.1.6  | DNS      | 92 Standard query 0x3c74 AAAA fa000000175.resources.office.net |
| 13719 | 26.441236 | 172.16.41.135 | 172.31.1.6  | DNS      | 76 Standard query 0x3ee1 AAAA fs.microsoft.com                 |

|                                                                                                                                                        |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Wireshark - Packet 45655 - Wi-Fi                                                                                                                       |  |
| ▶ Frame 45655: Packet, 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0 |  |
| ▼ Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)                                                     |  |
| ▶ Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)                                                                                                   |  |
| ▶ Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60)                                                                                                           |  |
| Type: IPv4 (0x0800)                                                                                                                                    |  |
| [Stream index: 4]                                                                                                                                      |  |
| ▼ Internet Protocol Version 4, Src: 172.16.41.135, Dst: 172.31.1.6                                                                                     |  |
| 0100 .... = Version: 4                                                                                                                                 |  |
| .... 0101 = Header Length: 20 bytes (5)                                                                                                                |  |
| ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)                                                                                        |  |
| Total Length: 78                                                                                                                                       |  |
| Identification: 0xffd5 (65493)                                                                                                                         |  |
| ▶ 000. .... = Flags: 0x0                                                                                                                               |  |
| ...0 0000 0000 0000 = Fragment Offset: 0                                                                                                               |  |
| Time to Live: 128                                                                                                                                      |  |
| Protocol: UDP (17)                                                                                                                                     |  |
| Header Checksum: 0x0000 [validation disabled]                                                                                                          |  |
| [Header checksum status: Unverified]                                                                                                                   |  |
| Source Address: 172.16.41.135                                                                                                                          |  |
| Destination Address: 172.31.1.6                                                                                                                        |  |
| [Stream index: 176]                                                                                                                                    |  |
| ▼ User Datagram Protocol, Src Port: 64655, Dst Port: 53                                                                                                |  |
| Source Port: 64655                                                                                                                                     |  |
| Destination Port: 53                                                                                                                                   |  |
| Length: 58                                                                                                                                             |  |
| Checksum: 0x8308 [unverified]                                                                                                                          |  |
| [Checksum Status: Unverified]                                                                                                                          |  |
| [Stream index: 539]                                                                                                                                    |  |
| [Stream Packet Number: 23]                                                                                                                             |  |
| ▶ [Timestamps]                                                                                                                                         |  |
| UDP payload (50 bytes)                                                                                                                                 |  |
| ▶ Domain Name System (query)                                                                                                                           |  |

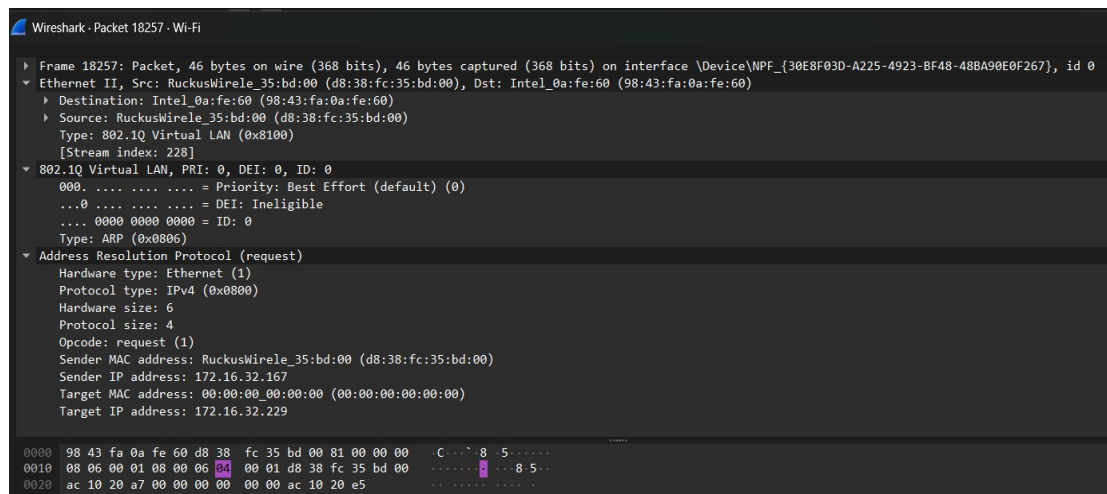
## 5. To capture and analyze HTTP packets

| File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help                                       |           |               |               |          |                    |
|------------------------------------------------------------------------------------------------------------------|-----------|---------------|---------------|----------|--------------------|
| http                                                                                                             |           |               |               |          |                    |
| No.                                                                                                              | Time      | Source        | Destination   | Protocol | Length Info        |
| 23790                                                                                                            | 53.450858 | 172.16.41.135 | 34.223.124.45 | HTTP     | 493 GET / HTTP/1.1 |
| Frame 23790: Packet, 493 bytes on wire (3944 bits), 493 bytes captured (3944 bits) on interface \Device\NPF{...} |           |               |               |          |                    |
| Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)                 |           |               |               |          |                    |
| Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)                                                               |           |               |               |          |                    |
| Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60)                                                                       |           |               |               |          |                    |
| Type: IPv4 (0x0800)                                                                                              |           |               |               |          |                    |
| [Stream index: 6]                                                                                                |           |               |               |          |                    |
| Internet Protocol Version 4, Src: 172.16.41.135, Dst: 34.223.124.45                                              |           |               |               |          |                    |
| 0100 .... = Version: 4                                                                                           |           |               |               |          |                    |
| .... 0101 = Header Length: 20 bytes (5)                                                                          |           |               |               |          |                    |
| Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)                                                    |           |               |               |          |                    |
| Total Length: 479                                                                                                |           |               |               |          |                    |
| Identification: 0x8662 (34402)                                                                                   |           |               |               |          |                    |
| 010. .... = Flags: 0x2, Don't fragment                                                                           |           |               |               |          |                    |
| ...0 0000 0000 0000 = Fragment Offset: 0                                                                         |           |               |               |          |                    |
| Time to live: 128                                                                                                |           |               |               |          |                    |
| Protocol: TCP (6)                                                                                                |           |               |               |          |                    |
| Header Checksum: 0x0000 [validation disabled]                                                                    |           |               |               |          |                    |
| [Header checksum status: Unverified]                                                                             |           |               |               |          |                    |
| Source Address: 172.16.41.135                                                                                    |           |               |               |          |                    |
| Destination Address: 34.223.124.45                                                                               |           |               |               |          |                    |
| [Stream index: 358]                                                                                              |           |               |               |          |                    |
| Transmission Control Protocol, Src Port: 55322, Dst Port: 80, Seq: 1, Ack: 1, Len: 439                           |           |               |               |          |                    |
| Hypertext Transfer Protocol                                                                                      |           |               |               |          |                    |

## 6. To analyze ARP request and reply packets



| No.    | Time       | Source                 | Destination       | Protocol | Length | Info                                      |
|--------|------------|------------------------|-------------------|----------|--------|-------------------------------------------|
| 5818   | 11.297199  | Intel_0a:fe:60         | Fortinet_6e:91:0f | ARP      | 42     | 172.16.41.135 is at 98:43:fa:0a:fe:60     |
| 5819   | 11.297227  | Intel_0a:fe:60         | Fortinet_6e:91:0f | ARP      | 42     | 172.16.41.135 is at 98:43:fa:0a:fe:60     |
| 35994  | 69.430284  | Fortinet_6e:91:0f      | Intel_0a:fe:60    | ARP      | 64     | Who has 172.16.32.19? Tell 172.16.32.1    |
| 4425   | 8.559894   | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.1? Tell 172.16.32.167   |
| 12753  | 24.701997  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.1? Tell 172.16.32.167   |
| 28969  | 54.727888  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.1? Tell 172.16.32.167   |
| 35975  | 69.275587  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.1? Tell 172.16.32.167   |
| 44039  | 84.753305  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.1? Tell 172.16.32.167   |
| 95972  | 114.769454 | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.1? Tell 172.16.32.167   |
| 103583 | 128.846928 | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.1? Tell 172.16.32.167   |
| 15400  | 29.354621  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.229? Tell 172.16.32.167 |
| 16488  | 30.357871  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.229? Tell 172.16.32.167 |
| 16959  | 31.358913  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.229? Tell 172.16.32.167 |
| 17619  | 32.361150  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.229? Tell 172.16.32.167 |
| 18257  | 33.362660  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.229? Tell 172.16.32.167 |
| 18732  | 34.365862  | RuckusWirele_35:bd:... | Intel_0a:fe:60    | ARP      | 46     | Who has 172.16.32.229? Tell 172.16.32.167 |
| 44186  | 85.025542  | Fortinet_6e:91:0f      | Intel_0a:fe:60    | ARP      | 64     | Who has 172.16.32.82? Tell 172.16.32.1    |
| 21987  | 40.777993  | Fortinet_6e:91:0f      | Intel_0a:fe:60    | ARP      | 64     | Who has 172.16.36.102? Tell 172.16.32.1   |
| 22643  | 41.778668  | Fortinet_6e:91:0f      | Intel_0a:fe:60    | ARP      | 64     | Who has 172.16.36.102? Tell 172.16.32.1   |
| 23066  | 42.778969  | Fortinet_6e:91:0f      | Intel_0a:fe:60    | ARP      | 64     | Who has 172.16.36.102? Tell 172.16.32.1   |
| 41616  | 80.473221  | Fortinet_6e:91:0f      | Intel_0a:fe:60    | ARP      | 64     | Who has 172.16.36.102? Tell 172.16.32.1   |



```
Wireshark - Packet 18257 - Wi-Fi
  Frame 18257: Packet, 46 bytes on wire (368 bits), 46 bytes captured (368 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0
    Ethernet II, Src: RuckusWirele_35:bd:00 (d8:38:fc:35:bd:00), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
      Destination: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
      Source: RuckusWirele_35:bd:00 (d8:38:fc:35:bd:00)
        Type: 802.1Q Virtual LAN (0x8100)
        [Stream index: 228]
      802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
        000. .... = Priority: Best Effort (default) (0)
        ...0 .... = DEI: Ineligible
        .... 0000 0000 0000 = ID: 0
        Type: ARP (0x0806)
      Address Resolution Protocol (request)
        Hardware type: Ethernet (1)
        Protocol type: IPv4 (0x0800)
        Hardware size: 6
        Protocol size: 4
        Opcode: request (1)
        Sender MAC address: RuckusWirele_35:bd:00 (d8:38:fc:35:bd:00)
        Sender IP address: 172.16.32.167
        Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)
        Target IP address: 172.16.32.229

0000  98 43 fa 0a fe 60 d8 38 fc 35 bd 00 81 00 00 00  -C... 8 5 .....
0010  08 06 00 01 08 00 06 04 00 01 d8 38 fc 35 bd 00  -..... 8 5 .....
0020  ac 10 20 a7 00 00 00 00 00 00 ac 10 20 e5  -.....
```

## 7. To identify source and destination MAC and IP addresses in captured packets.

The image shows a Wireshark packet capture analysis. The top pane displays a list of captured packets. The selected packet (No. 6892) is expanded, showing the Ethernet II and Internet Protocol Version 4 details.

| No.  | Time      | Source         | Destination    | Protocol | Length | Info                                                      |
|------|-----------|----------------|----------------|----------|--------|-----------------------------------------------------------|
| 6892 | 13.725480 | 172.16.41.135  | 13.204.106.34  | TCP      | 55     | 60059 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1               |
| 6896 | 13.728040 | 13.204.106.34  | 172.16.41.135  | TCP      | 66     | 443 → 60059 [ACK] Seq=1 Ack=2 Win=103 Len=0 SLE=1 SRE=2   |
| 7705 | 14.957805 | 172.16.41.135  | 74.125.200.188 | TCP      | 55     | 58464 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1              |
| 7746 | 15.038603 | 74.125.200.188 | 172.16.41.135  | TCP      | 66     | 5228 → 58464 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2 |

**Ethernet II Details:**

- Destination: Fortinet\_6e:91:0f (04:d5:90:6e:91:0f)
- Source: Intel\_0a:fe:60 (98:43:fa:0a:fe:60)
- Type: IPv4 (0x0800)
- [Stream index: 4]

**Internet Protocol Version 4 Details:**

- Version: 4
- Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 41
- Identification: 0xba38 (47672)
- Flags: 0x2, Don't fragment
- Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6)
- Header Checksum: 0x0000 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 172.16.41.135
- Destination Address: 13.204.106.34
- [Stream index: 213]

**Transmission Control Protocol Details:**

- Src Port: 60059, Dst Port: 443, Seq: 1, Ack: 1, Len: 1